

Sujith S

System Development Engineer | Java · Python · AWS · CI/CD · Cloud Infrastructure
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Summary

Computational Statistics and Data Analytics student at VIT University (2022-2027) with hands-on experience building, deploying, and automating production-grade backend systems on AWS. Proficient in Java and Python; applied experience in REST API development, CI/CD pipelines, Docker-based deployment, distributed data pipelines (Kafka, Spark), and PostgreSQL/Redis infrastructure. Track record of owning technical components end-to-end across the full SDLC, automating repetitive operational processes, troubleshooting root causes, and collaborating cross-functionally to improve system reliability, operational performance, and stability.

Education

Int.MSc Computational Statistics and Data Analytics

VIT University | 2022-2027

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Distributed Systems, Cloud Computing, Database Systems, Computer Networks, Machine Learning, Statistical Modeling

CS50 Programs

Harvard CS50 (Online)

Completed [CS50x](#), [CS50P](#), [CS50AI](#), [CS50SQL](#), and [CS50R](#) covering algorithms, full-stack development, SQL, Python, and AI fundamentals.

Skills

Languages: Java, Python, SQL, JavaScript, TypeScript, R

Cloud & Infrastructure: AWS (Lambda, S3, RDS, EKS, EventBridge), GCP, Azure, Docker, CI/CD Pipelines, Infrastructure Automation, Build/Release Automation

Backend & APIs: FastAPI, REST APIs, PostgreSQL, Redis, Apache Kafka, Apache Spark, Apache Airflow, dbt

Automation & DevOps: Deployment Automation, Pipeline Automation, MLflow, System Monitoring, Troubleshooting, Root Cause Analysis

ML & Data: Scikit-learn, XGBoost, LangChain, LangGraph, Hugging Face Transformers, Pandas, NumPy, Statistical Analysis

Projects

Social Media Sentiment Lakehouse

[GitHub](#)

Python, Apache Kafka, Apache Spark Streaming, Delta Lake, Hugging Face Transformers, Scikit-learn, MLflow, FastAPI, PostgreSQL, Docker, AWS

- Designed and deployed an end-to-end real-time streaming pipeline using Apache Kafka and Apache Spark Streaming for high-throughput distributed data ingestion and processing; managed Docker-containerized service deployments with PostgreSQL and Delta Lake sinks, maintaining operational stability and data consistency across components.
- Implemented automated NLP classification pipelines for real-time sentiment analysis at scale; built pipeline monitoring and failure alerting to enable automated recovery and reduce manual intervention; benchmarked model performance across classification approaches using MLflow.

Demand Forecasting System

[GitHub](#)

Python, Scikit-learn, XGBoost, Prophet, FastAPI, PostgreSQL, Redis, MLflow, Docker, AWS

- Architected and deployed production-grade REST APIs using FastAPI and Docker on AWS; designed PostgreSQL schema and Redis caching layer for high-concurrency request handling; implemented staging-to-production deployment workflows with automated validation checks, ensuring reliable system rollout.
- Built automated model retraining pipelines with drift detection, confidence interval estimation, and experiment tracking via MLflow; automated build, test, and release processes to eliminate manual deployment steps and improve operational reproducibility across model versions.

Autonomous Research Agent

[GitHub](#)

Python, LangChain, LangGraph, OpenAI APIs, FAISS, ChromaDB, FastAPI, PostgreSQL, Redis, Docker, AWS

- Designed and deployed a distributed multi-service system using FastAPI, PostgreSQL, Redis, and Docker on AWS; orchestrated modular service components with LangGraph for stateful multi-step execution; owned component-level design, implementation, testing, and production deployment end-to-end.
- Developed REST API endpoints integrating vector search (FAISS, ChromaDB) for knowledge retrieval; automated evaluation and performance benchmarking pipelines; documented operational procedures and system performance metrics to support cross-team knowledge sharing and iterative system improvement.

Work Experience

Founder

[Shoot the Breeze](#) | 2023 - Present

- Owned end-to-end SDLC for backend applications deployed on AWS: drove system design, implemented backend services in Python, managed Docker-based deployments across staging and production environments, and automated operational monitoring and incident response workflows; communicated technical trade-offs and status to cross-functional stakeholders.

App/Web Developer

[CropSky](#) | VIT-TBI Research & Innovation Hub | 2024 - Present

- Developed and deployed scalable REST APIs and backend systems at a university-incubated research startup; implemented CI/CD pipelines and automated build and release processes to improve deployment reliability and reduce manual overhead; collaborated cross-functionally with engineers and researchers to translate operational requirements into production-ready system components.

Certifications & Achievements

- [Google Generative AI Leader](#)
- [Google Cloud Computing Foundations](#)
- [FreeCodeCamp Legacy Full Stack](#)
- [OCI 2025 Certified Developer Professional](#)
- AWS [Solutions Architect](#) and [Developer](#)
- [Software Engineer Certification - HackerRank](#)
- Published a [research paper](#) on [stock market analysis and prediction](#) in the 14th proceedings of the [El Ruha International Conference on Scientific Research](#), pp. 730-740, 2025.